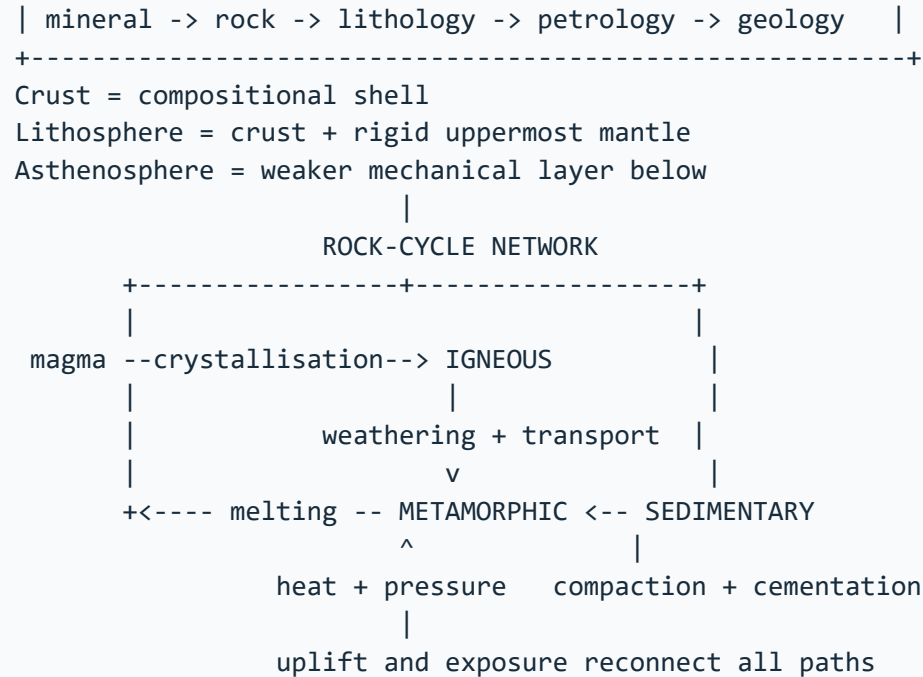


ASCII MASTER FLOW – PANEL 1/8: Crust, lithosphere and the rock-cycle network

EARTH MATERIAL SYSTEM



The cycle is a network: no rock family has one compulsory next stage.

ASCII MASTER FLOW – PANEL 2/8: Igneous, sedimentary and metamorphic diagnostic matrix

FAMILY	PROCESS	FIELD CLUES
Igneous	magma/lava cooling	crystals, glass, vesicles
intrusive	slow cooling	coarse texture
extrusive	fast cooling	fine or glassy texture
Sedimentary	deposit + lithify	beds, fossils, ripples
Metamorphic	solid-state change	grade, new minerals
foliated	differential stress	aligned fabric

Composition pairs:

granite-rhyolite = felsic | diorite-andesite = intermediate

gabbro-basalt = mafic

FIREWALLS

bedding is not foliation | loess and moraine are deposits until lithified

porosity is void space | permeability is connected flow capacity

WEGENER'S DRIFT CLAIM

ridge creation -> magnetic stripes -> older/deeper floor outward

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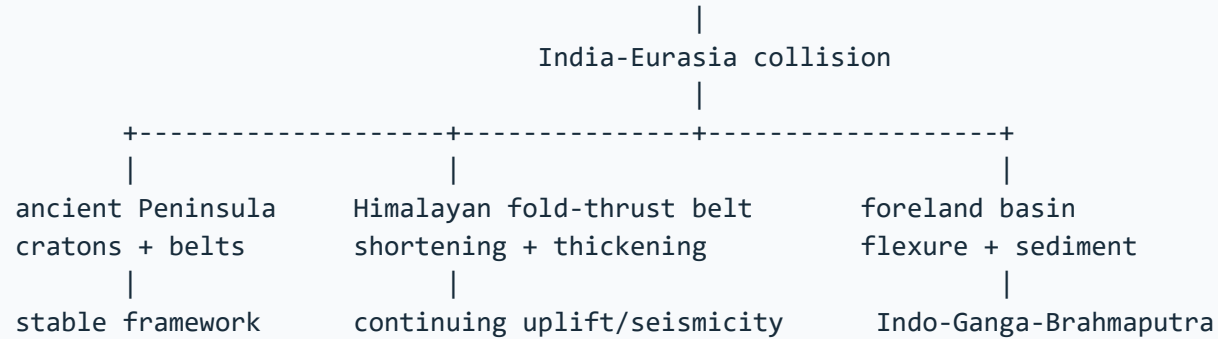
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Wilson cycle is an analogy, not a compulsory destiny for every ocean.

1

ASCII MASTER FLOW – PANEL 4/8: Indian Plate journey and the three-part geological skeleton

Indian rifting → northward Indian Plate drift → Tethys closure



NORTH-SOUTH EXAM CROSS-SECTION

Tethyan / Greater / Lesser / Siwalik Himalaya → foreland alluvium
→ Peninsular cratons, rifts, basins and Deccan cover

EVIDENCE LADDER

superposition → cross-cutting → unconformity → fossil correlation
→ radiometric event age → palaeomagnetism → geophysics

Stable Peninsula means relatively stable, not uniform, flat or aseismic.

ASCII MASTER FLOW – PANEL 5/8: Indian rock systems, basins and resource associations

SYSTEM	INDIAN EXPRESSION	SAFE ASSOCIATION	
+-----+	+-----+	+-----+	+
Archaean/Dharwar	gneiss + supracrustals	process-specific ore	
Proterozoic	Cuddapah, Vindhyan	stone; bounded ores	
Gondwana	faulted inland basins	coal, not universal	
Marine basins	coasts and offshore	petroleum potential	
Deccan Trap	stacked flood basalts	soil, stone, water	
+-----+	+-----+	+-----+	+

GONDWANA BASINS: Damodar | Son-Mahanadi | Satpura | Godavari

PETROLEUM SYSTEM:

source -> maturation -> migration -> reservoir -> seal -> trap -> timing

DECCAN ARCHITECTURE:

flow top / vesicles / fractures / weathered contact / dense flow interior

Association is not determinism: grade, preservation, exploration,
technology, economics and environmental limits decide usable resources.

ASCII MASTER FLOW – PANEL 6/8: Himalaya, foreland, coasts, islands and structural controls

HIMALAYA

collision orogen: HFT/MFT -> MBT -> MCT -> suture framework

|

FORELAND FLEXURE AND FILL

Bhabar -> Terai -> bhangar/khadar alluvial expressions

|

+-----+-----+	
PENINSULAR STRUCTURES	COASTS AND ISLANDS
Narmada-Son and Tapi rifts	west: rifted-margin sectors
Cambay and Kachchh	east: deltaic sediment basins
Godavari rift basin	Andaman: active island arc
joints, faults, bedding	Lakshadweep: coral atolls
+-----+-----+	

|

structure guides relief, drainage, waterfalls and groundwater

|

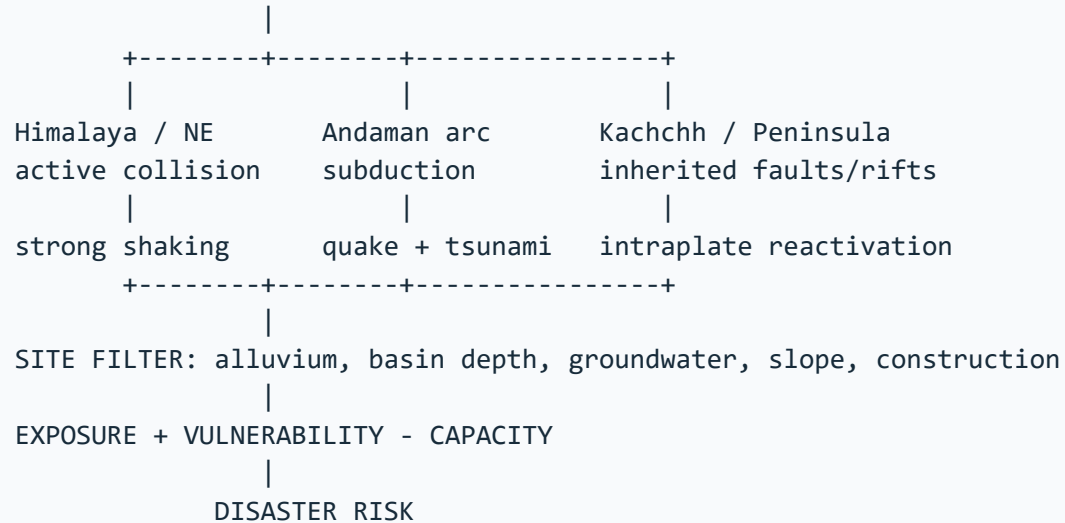
climate + erosion + base level + human action modify expression

Rift = extensional zone | graben = down-dropped block

Lineament = mapped linear feature, not automatically an active fault.

ASCII MASTER FLOW – PANEL 7/8: India seismic expression and geology-to-risk translation

TECTONIC AND INHERITED STRUCTURE



BIS Zones II-V are broad design generalisations, not predictions.
Geological association must be followed by site evidence and engineering.

RESPONSE: map -> investigate -> code -> inspect -> retrofit -> prepare.

ASCII MASTER FLOW – PANEL 8/8: Map-answer and Mains spine for India's geological architecture

QUESTION: How does geological structure shape India's surface geography?

1. DEFINE: age + composition + rock architecture + tectonic structures
2. DRAW: Himalaya -> foreland plain -> Peninsular shield cross-section
3. SEQUENCE: Gondwana rift -> drift -> collision -> uplift -> sediment fill
4. MAP NAMED UNITS
cratons | Purana basins | Gondwana basins | Deccan | coasts | islands
5. LINK PROCESS TO CONSEQUENCE
structure -> relief/drainage/water/resource/seismic expression
6. QUALIFY
association is not reserve; stable is not aseismic; alluvium has structure
7. VERDICT
inherited geology is filtered through tectonics, climate, erosion and use

Answer order: thesis -> map -> evidence -> mechanism -> India example
-> limitation -> graded conclusion.